

Name David Alvarado

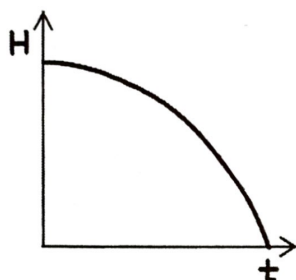
4

ChE 311 Quiz 4, Fall 2019

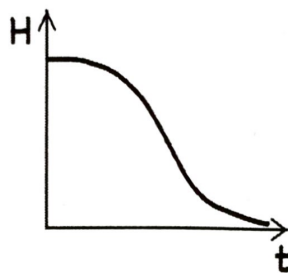
Closed book and closed notes, 10 minutes, 5 points

1. Choose one of the following answer that best represents the height of the liquid level as a function of draining time for the following cases.

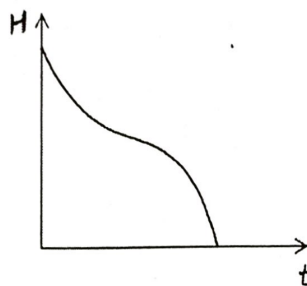
(A)



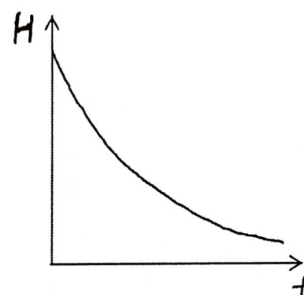
(B)



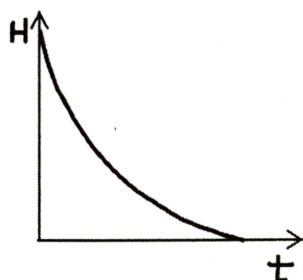
(C)



(D)



(E)



(F) Cannot decide without the specific dimensions of the tanks.



1) Draining of a cylindrical tank with a horizontal exit pipe (Figure 1.1). Answer: D

2) Draining of a cylindrical tank with a vertical exit pipe (Figure 1.2). Answer: E

3) Draining of a conical tank with a vertical exit pipe (Figure 1.3). Answer: A

4) Draining of a spherical tank with a vertical exit pipe. Initially the tank was half full (Figure 1.4). Answer: B

5) Draining of a spherical tank with a vertical exit pipe. Initially the tank was full (Figure 1.4). Answer: C

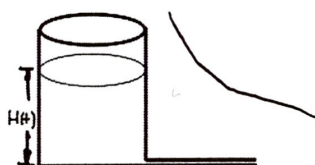


Figure 1.1

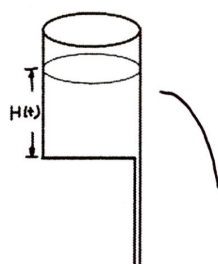


Figure 1.2

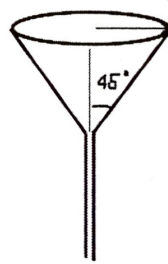


Figure 1.3



Figure 1.4

$$\frac{dm}{dt} = -\rho Q_2 \quad \rho_a - \rho_a g h$$

$$= -\rho \pi R^2 h_p^4$$

$$\frac{dm}{dt} = \frac{\rho \pi R^2 h_p^4}{8HL}$$

$$\rho_a - \rho_a g L$$

$$g h + g L$$

$$\rho g (h+L)$$

1

2

3

4, 5